Restricted Marginal Divergence: an efficient Bayesian measure of practical identifiability for nonlinear systems in biology and epidemiology

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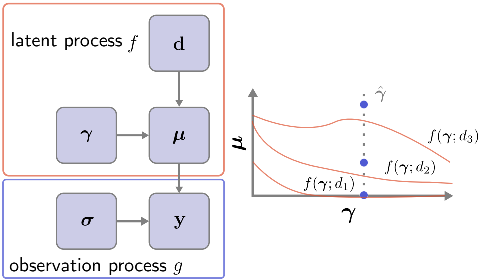
Nonlinear models

$$\mathbf{y} \sim g(\boldsymbol{\mu}, \boldsymbol{\sigma}) \qquad \text{observation process}$$

$$\boldsymbol{\mu} = f(\gamma; \mathbf{d}) \qquad \text{latent process}$$

- Model input $\mathbf{d}$ is called the design (measurement times, initial conditions, ...)
- Nonlinearity $\Rightarrow f$ is nonlinear in γ



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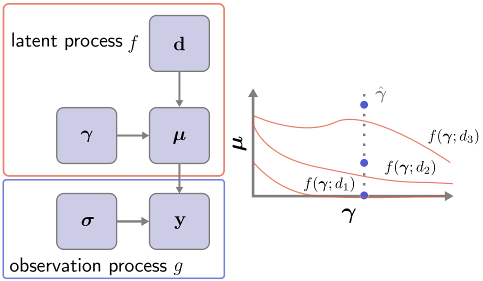

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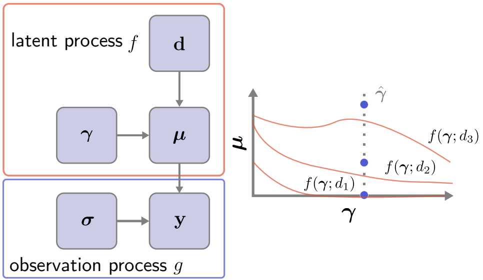
- More generally, interested some summary variable $u = \varphi(\boldsymbol{\theta})$



Model inference

- Together, f and g induce a density $P(\mathbf{y} \mid \boldsymbol{\theta})$ called a likelihood
  - Useful to think in terms of parameters : $\mathcal{L}(\boldsymbol{\theta}) = P(\mathbf{y} \mid \boldsymbol{\theta})$

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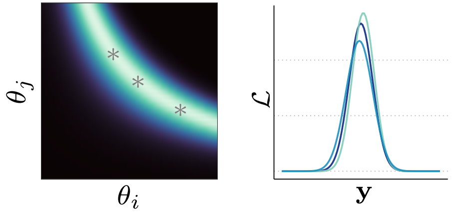

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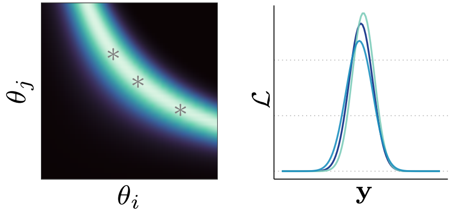
- Likelihood surface depends on the design, the true parameters, and the data actually observed



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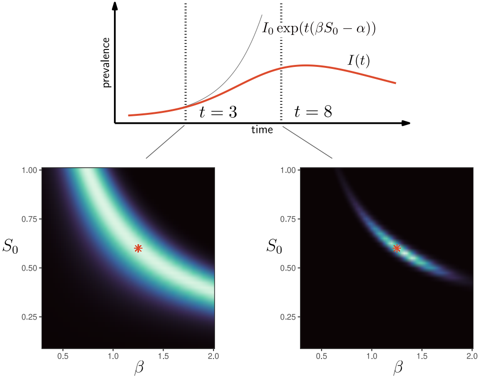
Example : SIR Model

$$\frac{d}{dt}S = -\beta SI \;, \quad \frac{d}{dt}I = \beta SI - \alpha I \;, \quad \text{and} \quad \frac{d}{dt}R = \alpha I$$

- Have $\mathbf{d} = (t_1, \ldots, t_n)^\top$, $\boldsymbol{\theta} = (\alpha, \beta, S_0)^\top$, and $\mu_t = I(t; \boldsymbol{\theta})$,
- Assuming (known) testing at rate η, have positive case counts

$$y_t \sim \text{Poisson}(\eta I(t; \boldsymbol{\theta}))$$

- Linear approximation shows exchangeability along $\beta S_0 - \alpha = \beta^* S_0^* - \alpha^*$

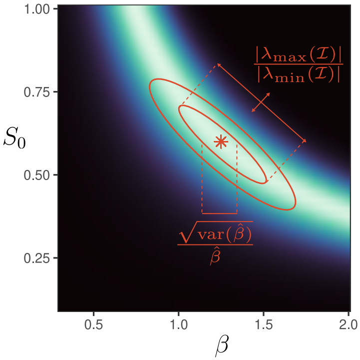


Existing approaches to PI

- Traditionally has focused on the covariance matrix of $\hat{\boldsymbol{\theta}}_{\text{MLE}} = \operatorname{argmax}_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$
- Under typical conditions, $\operatorname{cov}(\hat{\boldsymbol{\theta}}_{\text{MLE}}) \to \mathcal{I}(\boldsymbol{\theta}^*)^{-1}$, where

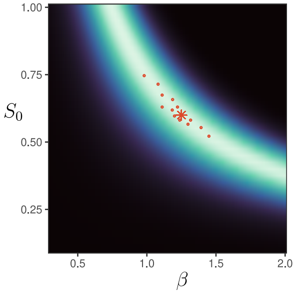
$$[\mathcal{I}(\boldsymbol{\theta})]_{ij} = -\mathbf{E}_{\mathbf{y}|\boldsymbol{\theta}}\left[\frac{\partial^2}{\partial\theta_i\partial\theta_j}\log\mathcal{L}(\boldsymbol{\theta})\right]$$

- PI commonly defined in terms of each coefficient of variation or the condition number



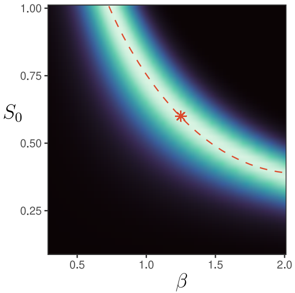
Other frequentist approaches

Monte Carlo method



- More accurate approximation of $\text{cov}(\hat{\boldsymbol{\theta}}_{\text{MLE}})$ in small data settings
- Sensitive to optimization procedure

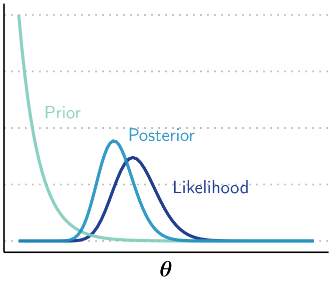
Profile likelihood



- Great for exploring likelihood surface
- Depends only on a single dataset
- Mainly qualitative

- Model uncertainty encoded through the *posterior distribution*

$$P(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{P(\mathbf{y} \mid \boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\mathbf{y})} \propto \mathcal{L}(\boldsymbol{\theta})P(\boldsymbol{\theta})$$



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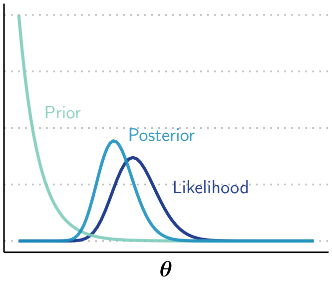

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$$P(u \mid \mathbf{y}) = \frac{P(\mathbf{y} \mid u)\, P(u)}{P(\mathbf{y})}$$

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- For transformations $u = \varphi(\boldsymbol{\theta})$, can be shown that

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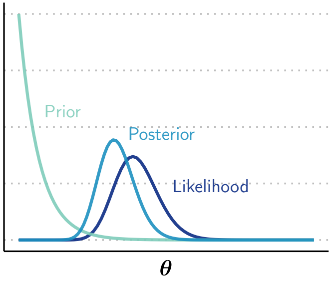

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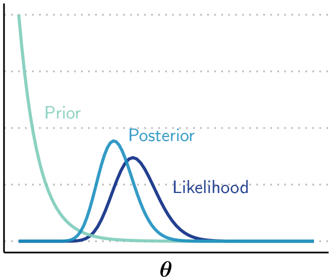

▷ $P(\mathbf{y} \mid u)$ is the evidence for a reduced model, with a prior $P(\boldsymbol{\theta} \mid u)$ restricted to dynamics matching u

For true parameters $\boldsymbol{\theta}^*$ and variable of interest $u = \varphi(\boldsymbol{\theta})$, the Restricted Marginal Divergence is defined

$$\mathrm{RMD}_u(\boldsymbol{\theta}^*) = \mathbf{E}_{\mathbf{y}|\boldsymbol{\theta}^*}\left[\log P(u^* \mid \mathbf{y})\right] - \log P(u^*)$$

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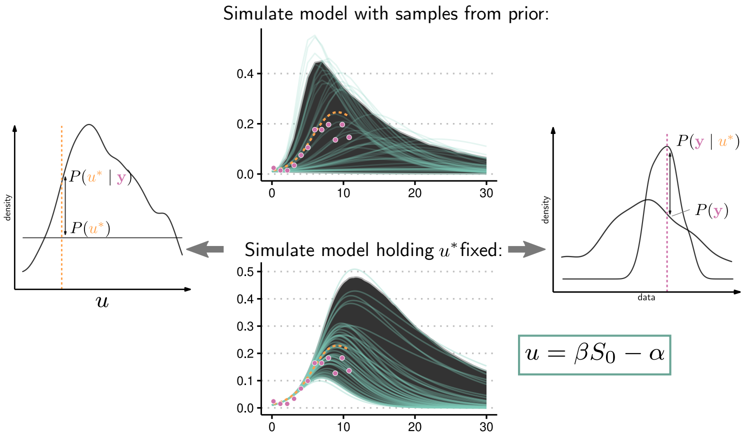
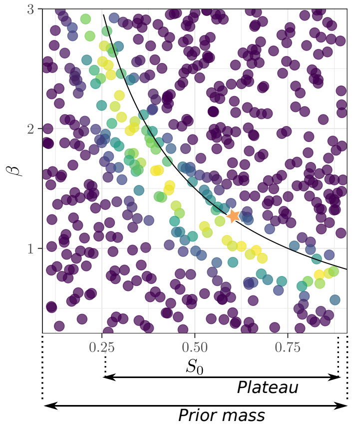
Parameter space $\longleftarrow \dfrac{P(u|\mathbf{y})}{P(u)} = \dfrac{P(\mathbf{y}|u)}{P(\mathbf{y})} \longrightarrow$ ***Data space***



RMD is a measure of *shrinkage* : it projects the global curvature of the likelihood onto u through marginalization



- **Asymptotics :** RMD converges with sufficient data to a function of the usual standard error approximation for $\hat{u}_{\text{MLE}}$,

$$\text{RMD}_u(\boldsymbol{\theta}^*) \approx \frac{1}{2}\left(\log n_t - \log \hat{\sigma}_u^2 - \log(2\pi)\right) - \log P(u^*)$$

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RMD Properties

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Have $\hat{\sigma}_u^2 = \mathcal{I}(\mathbf{v})_{ii}^{-1}$, i.e. the information matrix under a reparameterization $\mathbf{v} = (\theta_1, \ldots, \theta_{i-1}, u, \theta_{i+1}, \ldots, \theta_p)$,

$$\mathcal{I}(\mathbf{v}) = \left[\frac{\partial \boldsymbol{\theta}}{\partial \mathbf{v}}\right]^{-1} \mathcal{I}(\boldsymbol{\theta}) \left[\frac{\partial \boldsymbol{\theta}}{\partial \mathbf{v}}\right]$$

$$= \mathbf{V}^\top \mathbf{J}^\top \underbrace{\mathcal{O}(\boldsymbol{\mu})}_{\textit{curvature from } g} \overbrace{\mathbf{J}}^{\textit{curvature from } f} \underbrace{\mathbf{V}}_{\textit{curvature from } \varphi}$$

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- **Natural units :** definition in terms of densities means all variables on same scale. Penalizing by prior weight is similar to coefficient of variation
- **Global behavior :** assuming $\boldsymbol{\theta}^* \sim P(\boldsymbol{\theta})$, averaging $\text{RMD}_{\boldsymbol{\theta}}(\boldsymbol{\theta}^*)$ over $\boldsymbol{\theta}^*$ gives the mutual information $I(\mathbf{y};\boldsymbol{\theta})$, the amount of information about $\mathbf{y}$ that can be explained by the model [2]

2. Mattingly et al., "Maximizing the Information Learned from Finite Data Selects a Simple Model", *Proceedings of the National Academy of Sciences*. 2018.

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Attempt 1

- Sample datasets from $\mathbf{y}^{(i)} \sim P(\mathbf{y} \mid \boldsymbol{\theta}^*)$
- Obtain posterior samples from $P(u \mid \mathbf{y}^{(i)})$ using MCMC
- Approximate point density $P(u^* \mid \mathbf{y}^{(i)})$ and average over i

Calculating RMD

$$\mathrm{RMD}_u(\boldsymbol{\theta}^*) = \mathbf{E}_{\mathbf{y}|\boldsymbol{\theta}^*}\left[\log P(u^* \mid \mathbf{y})\right] - \log P(u^*)$$

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- Computationally expensive
- Requires density estimation (KLD)
- MCMC becomes sensitive when PI is in question

$$\mathbf{E}_{\mathbf{y}|\boldsymbol{\theta}^*}\left[\log \frac{P(\mathbf{y} \mid u^*)}{P(\mathbf{y})}\right] \quad \leftarrow \text{use this form instead!}$$

(Quasi) Monte Carlo

- Sample large set of parameters $\tilde{\boldsymbol{\theta}}^{(j)} \sim P(\boldsymbol{\theta} \mid u^*)$ and $\boldsymbol{\theta}^{(j)} \sim P(\boldsymbol{\theta})$
  - Independent, systematic samples seems to work well
- Sample datasets from $\mathbf{y}^{(i)} \sim P(\mathbf{y} \mid \boldsymbol{\theta}^*)$
- Average likelihoods $P(\mathbf{y}^{(i)} \mid \tilde{\boldsymbol{\theta}}^{(j)})$ and $P(\mathbf{y}^{(i)} \mid \boldsymbol{\theta}^{(j)})$ over j
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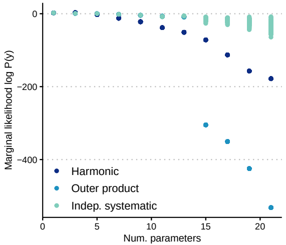

- Easy to parallelize
- The same bank of simulations can be resampled for each $\mathbf{y}^{(i)}$
- May become unstable with lots of data or many parameters

Sampling priors under the restricted model

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Instead, obtain samples using change of variables and Bayes' rule :

1. Choose a "pivot" parameter θ_i and define $\tilde{\varphi}(\theta_i \mid \boldsymbol{\theta}_{-i}) = u$
2. Obtain M samples from the target distribution

   $$P(\boldsymbol{\theta}_{-i} \mid u^*) \propto \prod_{j \neq i} P(\theta_j) P(u \mid \boldsymbol{\theta}_{-i})$$

   where

   $$P(u \mid \boldsymbol{\theta}_{-i}) = \left| \frac{d}{du} \tilde{\varphi}^{-1}(u \mid \boldsymbol{\theta}_{-i}) \right| P_{\theta_i}(\tilde{\varphi}^{-1}(u^* \mid \boldsymbol{\theta}_{-i}))$$

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▷ Upshot : users provide access to an inverse $\tilde{\varphi}^{-1}(u \mid \boldsymbol{\theta}_{-i})$ and its derivative

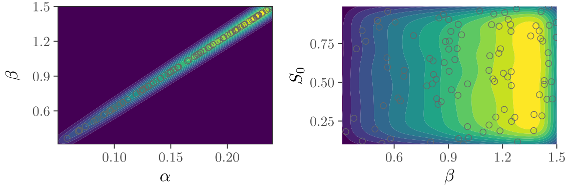
Example : basic reproductive number

Transformation : $\varphi(\boldsymbol{\theta}) = \frac{\beta}{\alpha} = \mathcal{R}$

1. Let β be the pivot : $\tilde{\varphi}(\beta \mid \alpha, S_0) = \mathcal{R}$
2. Solve

$$\tilde{\varphi}^{-1}(\mathcal{R} \mid \alpha, S_0) = \alpha\mathcal{R}$$
$$\frac{d}{d\mathcal{R}}\tilde{\varphi}^{-1}(\mathcal{R} \mid \alpha, S_0) = \alpha$$

3. Simulate samples $(\alpha^{(i)}, S_0^{(i)}) \sim P(\alpha, S_0 \mid \mathcal{R}^*)$ and plug in each $\beta^{(i)} = \mathcal{R}^*\alpha^{(i)}$



Application : accurately summarizing an outbreak takes time

Background

- Explosion of research in recent years applying epidemiological models to data [3]
- Statistics derived from these models are key to communication and public health response [4]
  - Inferences from even simple models are limited early in an epidemic [5]
  - Forces researchers to make strong assumptions about parameters or use retrospective data
- For SIR, previous work has mostly focused on PI of β and α with S_0 known [6, 7]

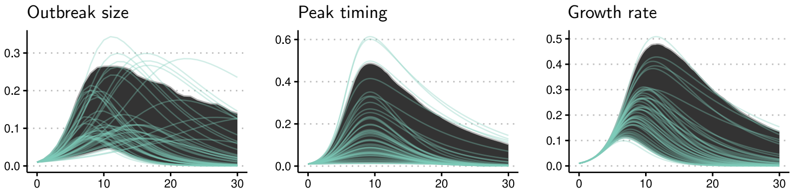
3. Chen, Allot et Lu, "LitCovid", *Nucleic Acids Research*. 2021.
4. Wu, K. Leung et G. M. Leung, "Nowcasting and Forecasting the Potential Domestic and International Spread of the 2019-nCoV Outbreak Originating in Wuhan, China", *Lancet (London, England)*. 2020.
5. Roda et al., "Why Is It Difficult to Accurately Predict the COVID-19 Epidemic ?", *Infectious Disease Modelling*. 2020.
6. Piazzola, Tamellini et Tempone, "A Note on Tools for Prediction under Uncertainty and Identifiability of SIR-like Dynamical Systems for Epidemiology", *Mathematical Biosciences*. 2021.
7. Tuncer et Le, "Structural and Practical Identifiability Analysis of Outbreak Models", *Mathematical Biosciences*. 2018.

1. How does PI early in an epidemic change when better accounting for limited data ?
2. Which statistics derived from models can we hope to learn most about, and when ?

Summary transformations used

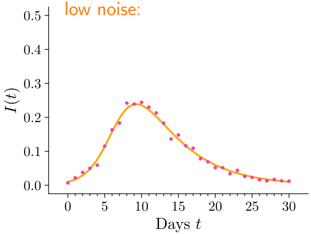
Name	Symbol	Formula
Reproductive number	$\mathcal{R}$	β/α
Outbreak size	$\mathcal{O}$	$1 - R(0) - S_0 \exp(-\mathcal{R}\mathcal{O})^*$
Peak intensity	$\mathcal{P}$	$I_0 + S_0 + [1 - \log(S_0/\mathcal{R})]/\mathcal{R}$
Peak timing	$\mathcal{T}$	Unknown
Growth rate	$\mathcal{G}$	$\beta S_0 - \alpha$

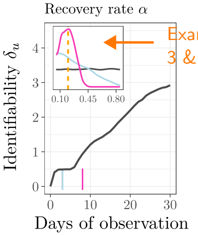
*Implicit equation



Experimental setup :

True outbreak with relatively low noise:

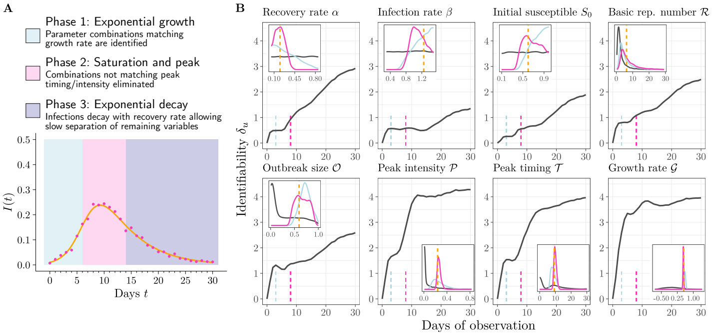




Example shrinkage after 3 & 8 days of data

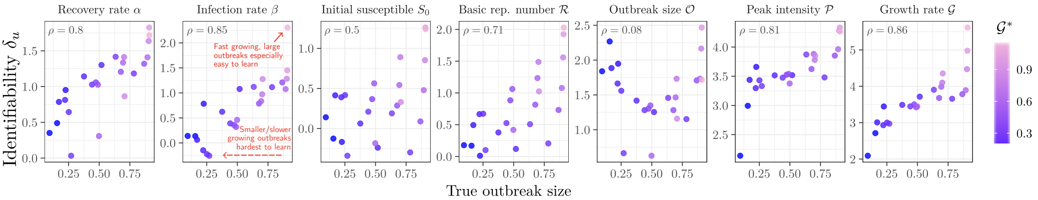
Each day = 1 additional observation

- Signature of 3 distinct stages of learning
- Sustained rate of learning following peak diverges from previous research [8, 9]



8. Capaldi et al., "Parameter Estimation and Uncertainty Quantication for an Epidemic Model", *Mathematical Biosciences and Engineering*. 2012.

9. Melikechi et al., "Limits of Epidemic Prediction Using SIR Models", *Journal of Mathematical Biology*. 2022.



- ○ Tested RMD over a grid of true values β^* and S_0^*
- ○ Observations up until true epidemic peak
- ▷ PI is especially challenging with less severe outbreaks, even with more observations

Acknowledgments

Collaborators :

Laurent Hébert-Dufresne

Jean-Gabriel Young

Julia package `MarginalDivergence.jl` : (Work in progress !)

Preprint for SIR Paper :

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